V66

Temporal root classification

Training loss

Validation loss

Shared reference B

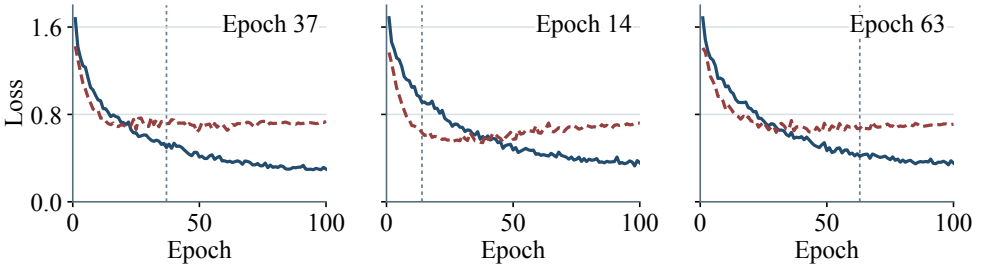
T1: baseline

Evaluation: T2–T4

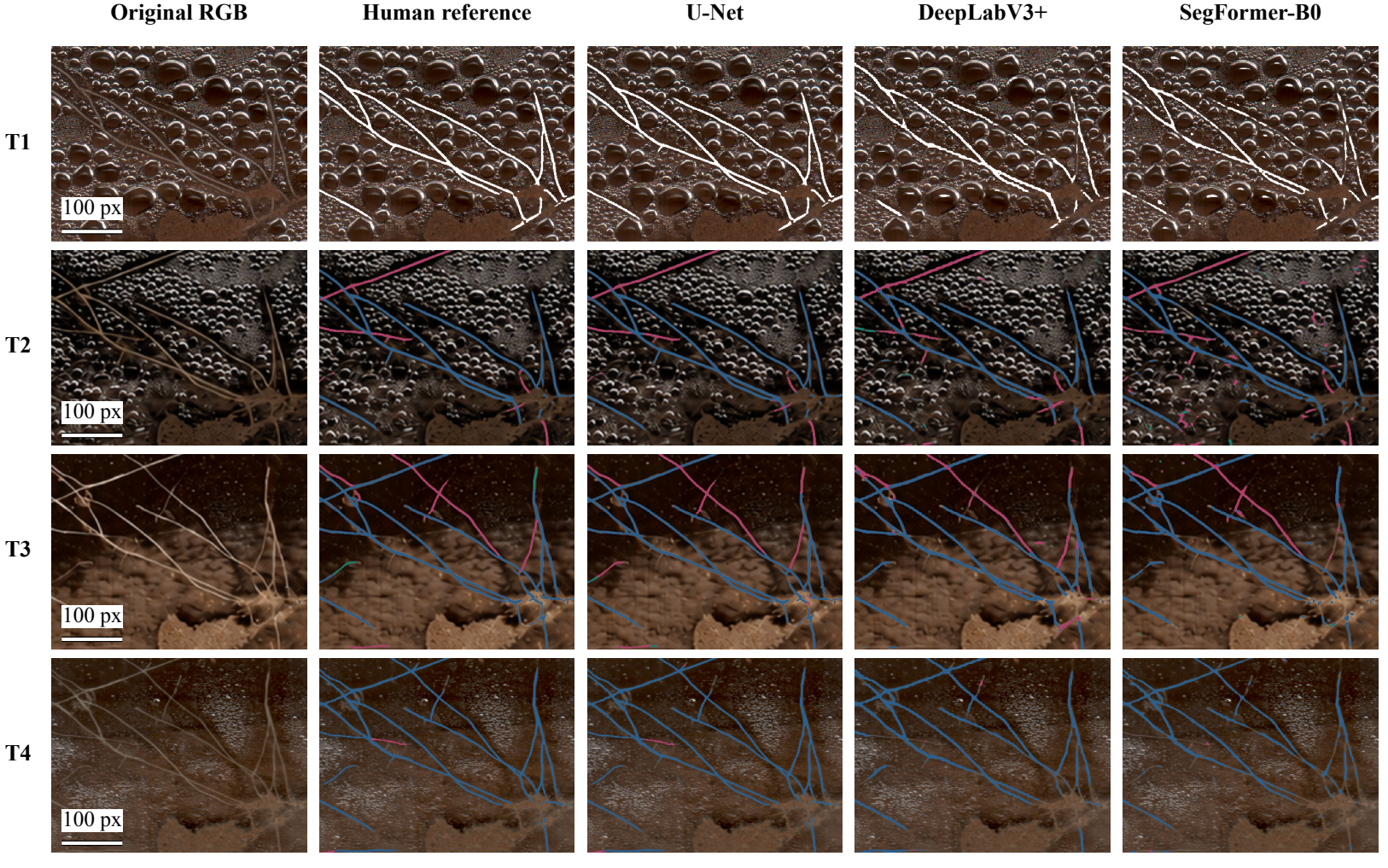
468,203 reference root pixels

Same local field across four dates

Baseline Ignored



Root class	U-Net Prec.	U-Net Recall	U-Net Dice	U-Net IoU	DeepLabV3+ Prec.	DeepLabV3+ Recall	DeepLabV3+ Dice	DeepLabV3+ IoU	SegFormer-B0 Prec.	SegFormer-B0 Recall	SegFormer-B0 Dice	SegFormer-B0 IoU
	Class-wise evaluation (%)				Class-wise evaluation (%)				Class-wise evaluation (%)			
Newly emerged	85.0	85.3	**85.2**	74.2	74.6	64.8	**69.3**	53.1	81.0	68.0	**73.9**	58.7
Elongating	26.2	38.7	**31.2**	18.5	19.9	17.6	**18.7**	10.3	33.5	18.3	**23.7**	13.4
Persistent	98.1	97.0	**97.6**	95.2	97.7	78.6	**87.1**	77.1	97.1	75.6	**85.0**	73.9
	Macro Dice: 71.32%				Macro Dice: 58.37%				Macro Dice: 60.89%			



Scale bars: 100 px. Colors denote temporal classes; reference B is shared across models.